Final Project

Sugarcane Yield Prediction

Group 9

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Abstract
Sugarcane cultivation is a key agricultural activity across South Asia, playing a critical role in supporting rural livelihoods, boosting economic development, and supplying raw materials to various industries. This study aims to analyze the South Asian Sugarcane Production dataset to uncover the primary factors affecting sugarcane yield and to develop predictive models that support optimized farming practices.
The dataset includes diverse features such as soil types, fertilizer compositions, irrigation techniques, cane varieties, climatic conditions, and yield grades, gathered from five South Asian countries over several decades. Leveraging statistical and machine learning approaches, the project focuses on building robust classification models to accurately predict yield category based on the contributing agricultural and environmental factors.
Our project is intended to assist agricultural stakeholders such as farmers, policymakers, and agronomists in making informed decisions to maximize productivity. By applying modern modeling techniques, we aim to provide actionable insights that promote sustainable farming and resource efficiency across the region.

"Sustainable agriculture feeds the world without draining its future."

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1. Introduction

Sugarcane is a vital crop in South Asia, contributing to the production of sugar, ethanol, and other by-products, and playing a key role in the economy. Predicting sugarcane yield, however, is challenging due to factors like soil quality, water availability, fertilizer use, and climate variations. In countries such as India, Bangladesh, Nepal, Pakistan, and Sri Lanka, issues like poor water management, diverse soil types, and inconsistent fertilizer practices, along with pests and fluctuating weather, complicate yield prediction.

To address this, our project aims to develop a statistical model that predicts sugarcane yields using a dataset that includes factors like soil type, water usage, fertilizer type, and environmental conditions. By applying machine learning techniques, we will identify the most significant factors impacting yield and improve prediction accuracy.

The model will categorize sugarcane yields into two groups (e.g., High and Low), offering actionable insights for farmers to optimize their practices. This data-driven approach aims to improve the sustainability and efficiency of sugarcane farming in South Asia.

2. Description of the Question

As sugarcane production plays a vital role in the economies of South Asia, optimizing its yield is critical for ensuring food security, improving farmer livelihoods, and supporting agricultural industries. Understanding the factors that influence sugarcane yield and predicting its outcomes are essential for better resource management and agricultural practices. Predictive models for sugarcane yield can help farmers make informed decisions regarding soil treatment, irrigation, and fertilizer use, leading to improved productivity. Therefore, we aim to understand:

- **What are the key factors that influence sugarcane yield?**
- **How can we develop a reliable statistical model to predict sugarcane yield based on these factors?**

By addressing these questions, we aim to create a predictive model that helps improve sugarcane farming practices, leading to sustainable growth and optimized yield in South Asia.

3. Description of the Dataset

The Sugarcane Yield Prediction dataset was taken from Kaggle. It contains a total of 50 000 observations and 20 variables. One of these variables is "Yield," which serves as the response variable for this study. The dataset includes various factors that influence sugarcane yield, such as soil type, water usage, fertilizer type, and environmental conditions. Below is a brief description of each variable:

Variable	Description
Quantitative Variables	
Area_Remain	The remaining cultivated area after yield harvesting.
Yield	The actual crop yield obtained
ContractsArea	The area under contract farming.
Rain_Avg	The average rainfall in the region.
Qualitative Variables	
Year	The year in which the farming data was recorded.
Country	The country where the farming activity took place (India, Bangladesh, Pakistan, Sri Lanka, Nepal).
ActionWater	The method of water application (e.g., Rain, Water Pour).
WaterType	The source of water used for irrigation (e.g., Rain, Groundwater, Natural Canal).

Class_cane	The classification of sugarcane growth stage (e.g., 1st ratoon, 2nd ratoon, 3rd ratoon).
Epidemic	Type of epidemic affecting the crop (e.g., Herbicide, Preemergent).
YieldOldGrade	Previous grade classification of the yield (Grade 1, Grade 2, Grade 3).
Type_Cane	The specific variety of sugarcane planted (e.g., K84-200, LK92-11, Khonkean).
GrooveWidth	The spacing between planting grooves (measured in cm, e.g., 120 cm, 150 cm).
FertilizerType	The type of fertilizer used (Organic or Chemical).
TypeSoil	The type of soil in which the crop was grown (e.g., Silty Clay, Loam, Ferus Soil).
Fertilizer	The specific fertilizer formula applied (e.g., 15-15-15, 16-16-16, 46-0-0).
TargetOldGrade	The target yield grade of the crop (e.g., Grade 1, Grade 2, Grade 3).
FarmerContractGrade	The grade of the farmer's contract (e.g., A, B, C).
YieldGrade	The grade of the yield (e.g., Grade 1, Grade 2, Grade 3).
TargetGrade	The expected grade classification before harvest.

Table 1 - Variable Description

4. Data Pre-processing

Missing Values: No missing values were found in the dataset.
Duplicates: No duplicate records were found in the dataset.

Data entry errors: During data cleaning, we identified an erroneous fertilizer entry, "25-07-2007," appearing in 12,473 records, about 1/5 of the dataset. This was likely a data entry mistake, as it doesn't correspond to a valid fertilizer type. Instead of removing these records, we replaced the incorrect value with the meaningful fertilizer type "25-7-7." This approach allowed us to retain the data while ensuring consistency and maintaining the dataset's integrity for analysis.

4.1. Feature Engineering

Recategorized Variables:
Yield (Target variable): numerical→ 2 categories (High, Low)

Previous	New
Yield	Yield_Category

- Year is categorized into 10 ranges for EDA and FAMD to uncover hidden pattern.
- Then, the dataset was split into training and test sets. The training set consisted of 40,000 observations. EDA and model fitting was conducted using this training set.

5. Main Results of Descriptive Analysis

5.1. Univariate Analysis

Distribution of Response Variable

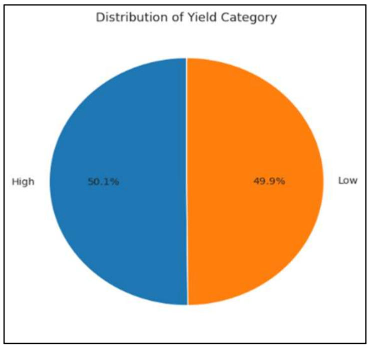
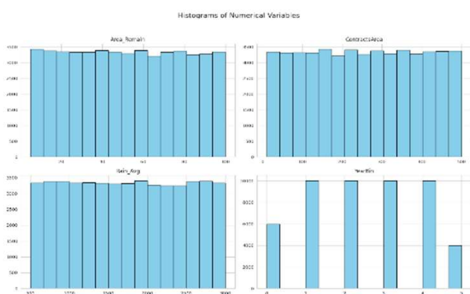


Figure 1 - Distribution of yield category

This plot shows the distribution of the target variable, "Yield Category," with two categories: "Low" and "High." The plot indicates that the dataset is balanced, with roughly equal counts for each category.



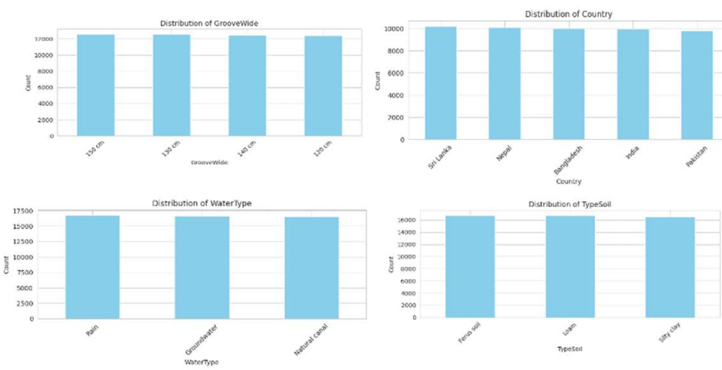
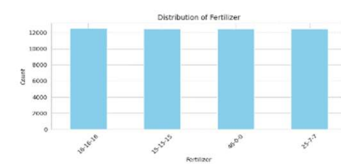


Figure 2 - Univariate graphs



- ➢ Here are the univariate plots for several categorical variables. All the plots show equal or nearly equal counts across each category, indicating that there is little to no variance between the categories for each variable.
- ➢ When a variable has no variance, It might not be helpful in predicting the target variable, because it's not showing any relationship or pattern that differentiates high vs. low yield.
- ➢ Since several categorical variables show the same kind of uniform distribution, some of them might be redundant and could be removed during feature selection to simplify the model.

5.2. Bivariate Analysis

Association between Yield Category vs Area Remain

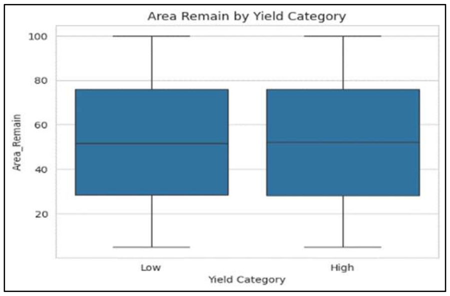


Figure 3 - Boxplot of Yield Category vs Area Remain

The median values for Area_Remain are similar for both yield categories. Both categories have a similar range, with values spanning from approximately 20 to 100.

The interquartile range (IQR) is slightly narrower for the "High" yield category compared to "Low," indicating less variability in Area_Remain for "High" yields.

Association between Yield Category vs Rain_Avg

The median values of Rain_Avg are very similar for both "Low" and "High" yield categories (around 1750).

The interquartile range (IQR) is also quite similar.

The range of rainfall values (indicated by the whiskers) is broad and nearly identical for both categories, spanning from around 500 to 3000.

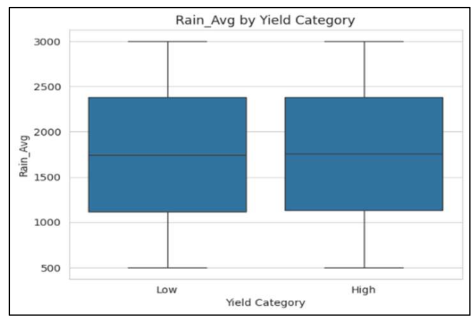


Figure 4 - Boxplot of Yield Category vs Rain_Avg

Association between Yield Category vs ContractsArea

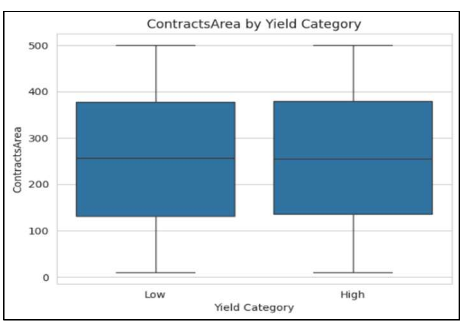


Figure 5 - Boxplot of Yield Category vs ContractsArea

The median values for ContractsArea are nearly identical for both "Low" and "High" yield categories, both around 250.

The interquartile ranges (IQRs) are also very similar, indicating similar variability in ContractsArea for both groups.

The ranges (indicated by the whiskers) are broad, from about 10 to 500, and nearly the same for both categories.

Association between Yield Category vs GrooveWide, Country, ActionWater and Fertilizer

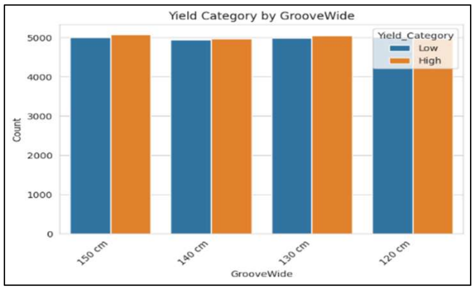


Figure 6 - Boxplot of Yield Category vs GrooveWide

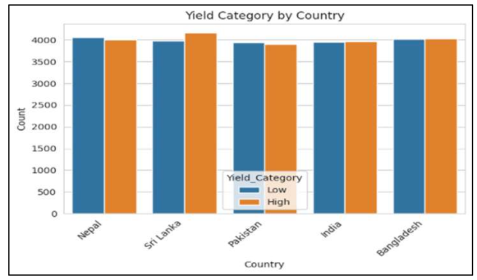


Figure 7 - Boxplot of Yield Category vs Country

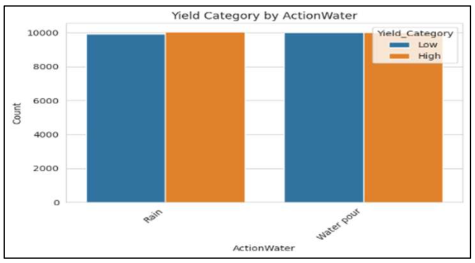


Figure 8 - Boxplot of Yield Category vs ActionWater

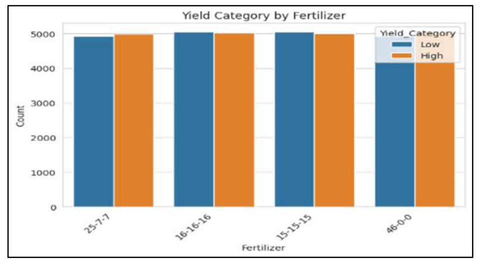


Figure 9 - Boxplot of Yield Category vs Fertilizer

The distribution of "Low" and "High" yield categories is almost uniform across different groove widths (150cm, 140cm, 130cm, 120cm), across all countries (Nepal, Sri Lanka, Pakistan, India, Bangladesh), both "Rain" and "Water pour" irrigation methods of Action Water variable and across various fertilizer types (25-7-7, 16-16-16, 15-15-15, 46-0-0). It suggests that all four variables may not strongly influence the yield category.

5.3. Correlation Analysis

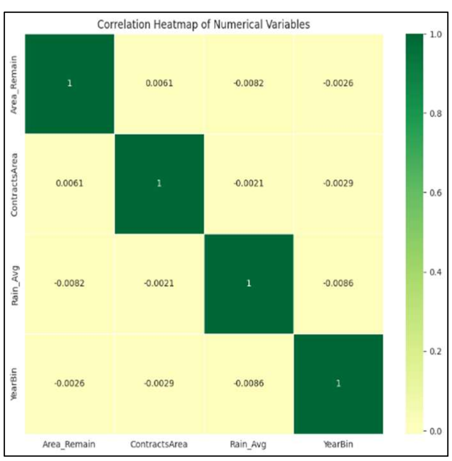


Figure 10 – Correlation Heatmap for Numerical Variables

Correlation heat map of numerical variables indicates that there are no significant correlations between numerical variables. Almost all the correlation coefficients between numerical variables are below 0.01 while the highest being 0.0061.

The numerical variables in this dataset are largely independent of each other. Changes in one variable are unlikely to have a significant linear impact on the.Models relying on linear relationships between these variables may not perform well. Other relationships or other variables may need to be considered.

5.4. Factor Analysis for Mixed data

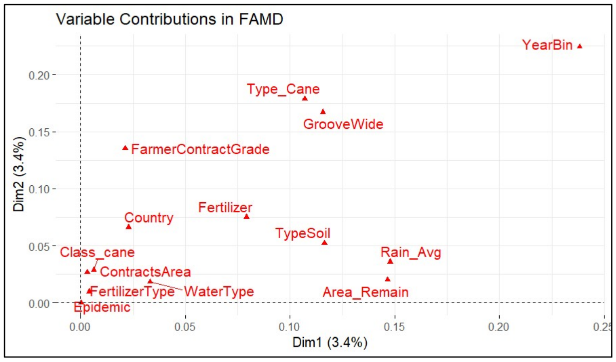


Figure 11 – Variable Contribution in FAMD

Variables farther from the origin contribute more to the variation explained by these dimensions. In here, YearBin, Type_Cane, and GrooveWide contribute notably to Dim1, while YearBin, Type_Cane, and FarmerContractGrade are influential for Dim2. In the factor map of variable contributions (Dim1 vs. Dim2), the variables Type_Cane and GrooveWide are positioned closely and point in a similar direction, both located relatively far from the origin. This spatial proximity and shared orientation suggest a positive correlation between these two variables within the dimensions captured by the analysis.

This plot showing how much each variable contributes to the first dimension (Dim1) in percentage. Variables like YearBin, Rain_Avg, Area_Remain, and TypeSoil have the highest contributions. The red dotted line marks the average expected contribution—variables above this line are more important for Dim1.

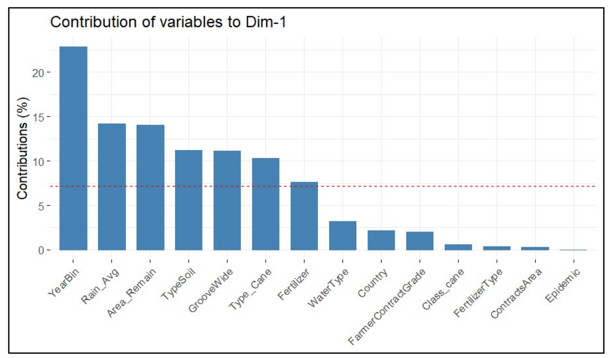


Figure 12 – Contribution of Variables to Dim-1

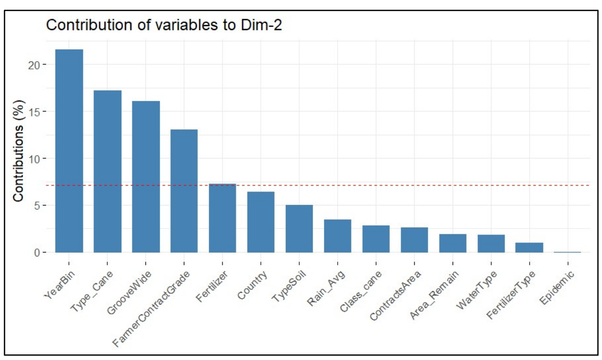


Figure 13 – Contribution of Variables to Dim-2

Top contributors include YearBin, Type_Cane, GrooveWide, and FarmerContractGrade.

The distance from the origin reflects the strength of correlation with the dimensions. Color represents the cos² value of the variables on the factor map. Rain_Avg and Area_Remain contributes more clearly, while the ContractsArea is closer to the center, meaning it is less well represented on Dim1 and Dim2. Perpendicular variables are uncorrelated. Therefore contractsArea and Area remain are uncorrelated.

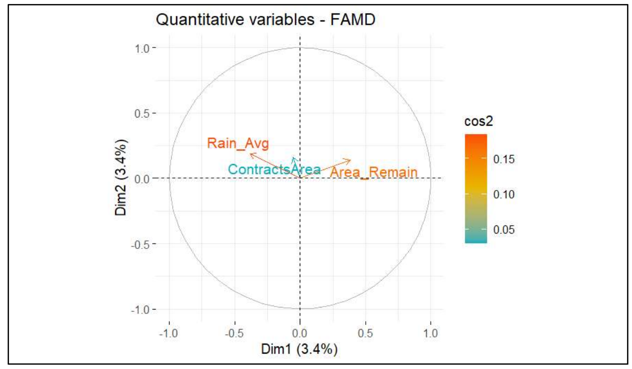


Figure 14 – Quantitative Variables - FAMD

5.5. Cluster Analysis with K-Prototype method.

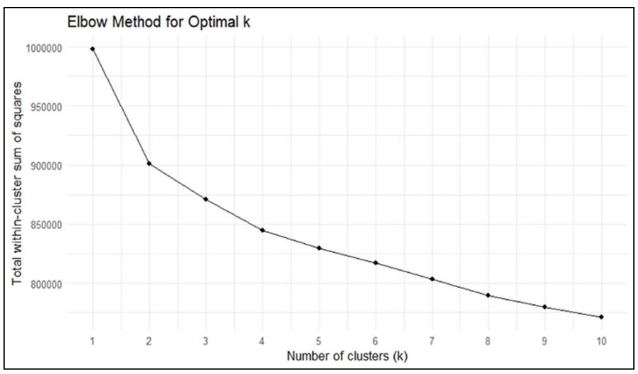


Figure 15 – Elbow Method for Optimal k

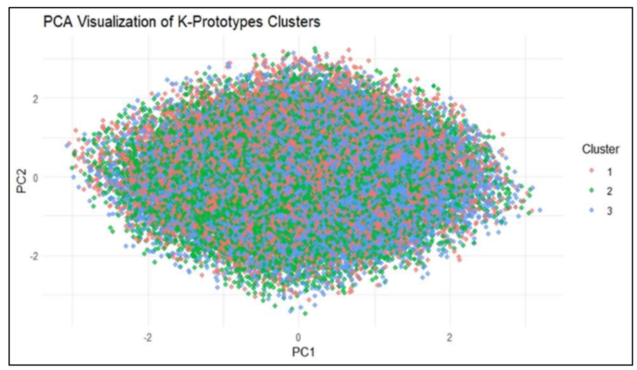


Figure 16 – PCA Visualization of K-Prototype Clusters

The K-Prototypes clustering algorithm was applied since our dataset contains both numerical and categorical variables. Identifying potential clusters is an important step for advanced analysis, as it may reveal hidden structures or patterns in the data.

To determine the optimal number of clusters, we experimented with different values of k using the Elbow Method. The elbow appeared around k = 2 and k = 3, with corresponding Silhouette Scores of 0.002 and -0.008, respectively. For k = 4, the Silhouette Score further decreased to -0.01. This trend—where the score remains near zero or negative as k increases—indicates that the clustering structure is weak, with significant overlap between clusters.

Although we intended to visualize the Elbow Method plot, the large size of the dataset made it computationally challenging to generate. However, based on the Silhouette Scores and visual inspection of the clusters in the PCA-reduced dimensions, it's clear that the clusters are highly overlapping. It's important to note that PCA is designed for numerical variables, so it may not fully capture separation based on categorical features. Overall, the near-zero Silhouette Scores suggest that no meaningful clusters exist in the data.

5.6. Outlier Analysis

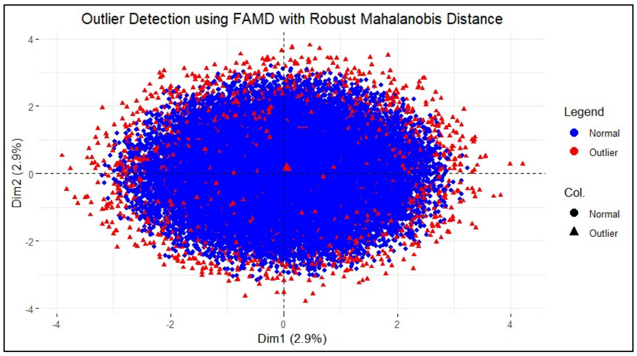


Figure 17 – Outlier Detection using FAMD with Robust Mahalanobis Distance

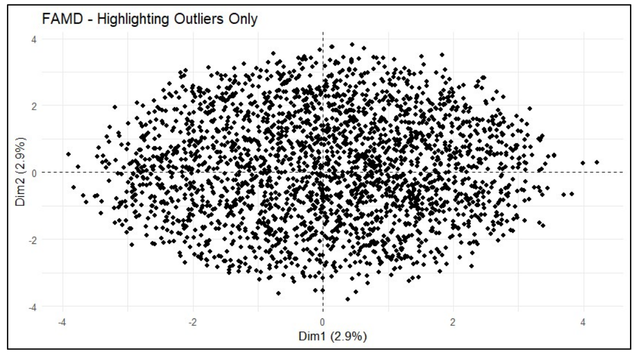


Figure 18 – FAMD-Highlight Outliers Only

To identify outliers, we applied two methods:

1. **Isolation Forest** was used on the numerical variables in the dataset. By selecting the top 5% of anomaly scores, we detected 2,500 outliers.
2. Additionally, we applied Factor Analysis of **Mixed Data (FAMD) followed by robust Mahalanobis distance** to accommodate the mixed nature of our dataset (containing both numerical and categorical variables). Using the top 5% of Mahalanobis distances, this method also identified 2,500 outliers.

FAMD was applied first due to the presence of mixed data types, ensuring that both numerical and categorical variables were fairly represented in the dimensionality reduction.

While approximately 6.25% of the dataset was flagged as outliers, we chose not to remove them, as there was no strong justification for excluding those data points. Instead, as a precaution, we conducted modeling both with and without the detected outliers. Interestingly, the models trained with outliers resulted in better accuracy and F1 scores, which led us to retain all data points for the advanced analysis.

The plot above visualizes the identified outliers in a 2D projection based on the first two FAMD dimensions, highlighting their distribution in the reduced feature space.

6. Main Results of Advance Analysis

6.1. Random Forest Model & Logistic Model

Feature Selection
Method: Mutual Information

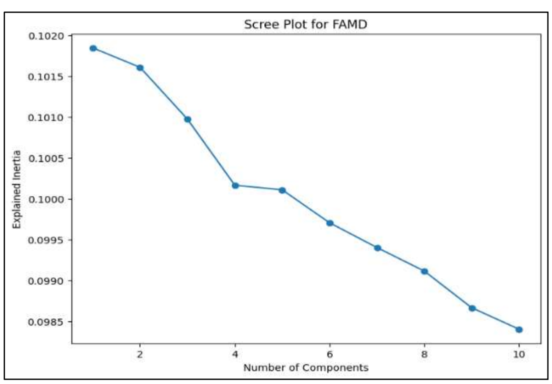
Feature	Mutual Information
Country	0.006680
Fertilizer Type	0.005755
Action Water	0.005049
Type Cane	0.003444
Epidemic	0.002917
Yield Old Grade	0.002574
Class cane	0.002365
Type Soil	0.002312
Target Old Grade	0.001929
Year	<0.0001
Groove Wide	<0.0001
Area Remain	<0.0001
Water Type	<0.0001
Rain_ Avg	<0.0001
Contracts Area	<0.0001
Fertilizer	<0.0001
Yield Grade	<0.0001
Farmer Contract Grade	<0.0001
Target Grade	<0.0001

Mutual Information is a robust method to measure the information that a feature is representing related to the target.

According to the mutual information scores almost all the representations of information are significantly small. Here the features with mutual information less than 0.0001 were considered insignificant. However, incorporating domain knowledge some of them were not removed assuming the existence of latent features or interacting terms. Thus, variables that did not show any relation to the yield level were removed.

Removed features: ContractsArea, Area_Remain, YieldGrade, TargetGrade, FarmerContractGrade.

Identifying Latent Features & Interacting Terms
Method: Factor Analysis for Mixed Data



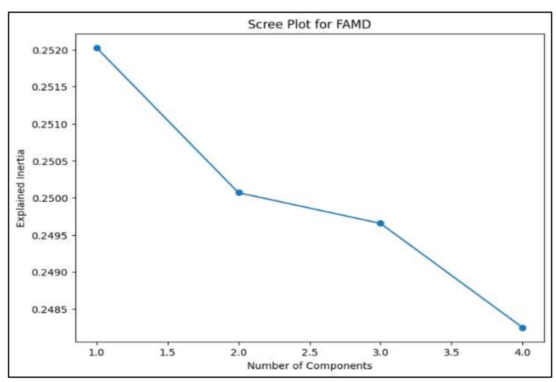


Figure 19 – Scree plot for FAMD

Elbow point was observed at n=4. Thus, four components were selected, the first two components representing approximately 50%+ explained variance.

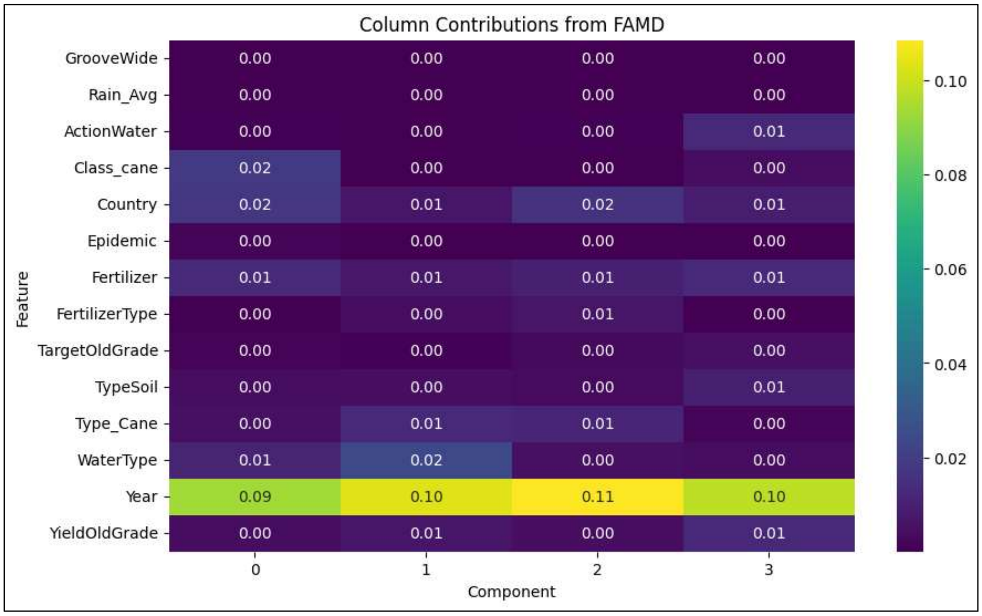


Figure 20 – Column Contribution for FAMD

Heatmap represents the contribution of each feature to the components (latent features). Year has a significant contribution in all the components indicating high season or cycle based fluctuations and the influence of technological farming transitioning into the modern era.

Proposed Latent Features

Component 1: Year to year trends connected with cane harvest ratoon and country.

Component 2: Year to year irrigation type differences.

Component 3: Regional treatment interaction over time. (Technological improvements included)

Component 4: Climate and weather based seasonal or cyclic trends.

Model Fitting (Random Forest) Before FAMD

Class	Precision	Recall	F1-Score
Low	0.51	0.58	0.54
High	0.49	0.42	0.45
Training Accuracy		0.5193	
Testing Accuracy		0.5017	

Table 2 - Model Fitting (Random Forest) Before FAMD

Model Fitting (Random Forest) After FAMD

Class	Precision	Recall	F1-Score
Low	0.52	0.51	0.51
High	0.50	0.51	0.50
Training Accuracy		0.5292	
Testing Accuracy		0.5077	

Table 3 - Model Fitting (Random Forest) After FAMD

After applying FAMD F1-Scores have improved. However there is no significant improvement in the overall accuracy.

Model Fitting (Logistic Model) After FAMD

Class	Precision	Recall	F1-Score
Low	0.51	0.57	0.54
High	0.50	0.44	0.47
Training Accuracy		0.5062	
Testing Accuracy		0.5049	

Table 4 - Model Fitting (Logistic Model) After FAMD

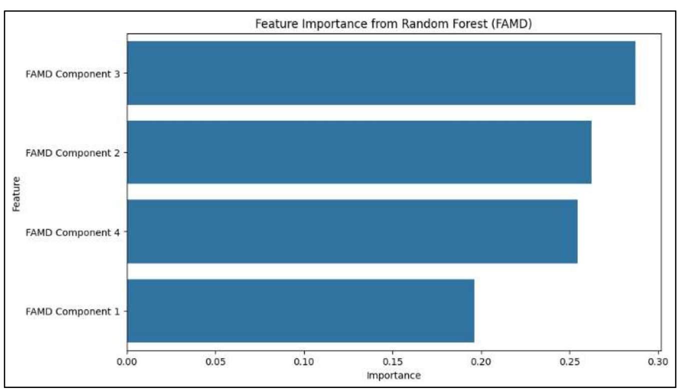


Figure 21 – Feature Importance for Random Forest

Latent Feature 3 has the highest importance in the random forest model and it is similar to the logistic model as well. Which indicates the model is focused more on treatments and irrigation related factors interacting with time.

6.1. XGBoost: Extreme Gradient Boosting

Overview

Extreme Gradient Boosting (XGBoost) is a high-performance ensemble learning algorithm that builds multiple decision trees to enhance predictive accuracy. It excels in handling large datasets, class imbalance, and multicollinearity.

Justification for Using XGBoost

Selected for its robust performance with large. Effectively manages non-linear feature interactions. Suited for our binary classification task: Low vs. High yield (50000 observations).

Model Performance

Metric	Train	Low Yield	High Yield	Test	Low Yield	High Yield
Accuracy	0.5755	-	-	0.5107	-	-
AUC-ROC	0.6084	-	-	0.4917	-	-
Precision	-	0.5875	0.5670	-	0.4999	0.5184
Recall	-	0.4885	0.6614	-	0.4235	0.5941
F1-score	-	0.5750	0.6105	-	0.4586	0.5536

Figure 22 – Model Performance for XGBoost

Training accuracy is 57.55% and Test accuracy is 51.07%. F1 scores are around 0.5 for training and test low and high yield categories.

Feature Selection and Insights

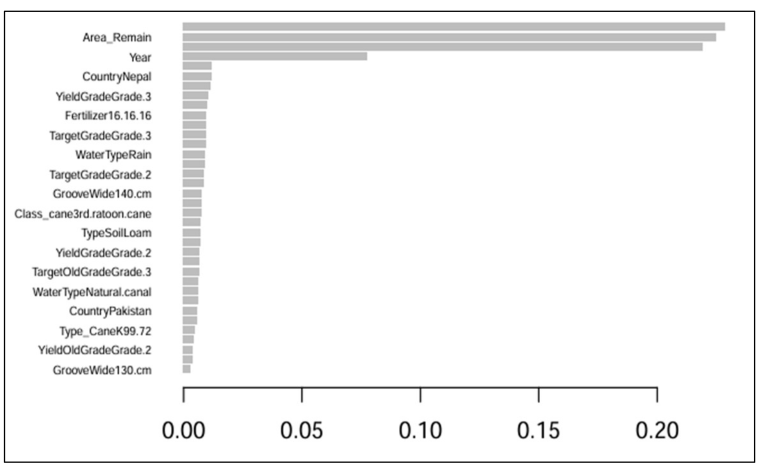


Figure 23 – Feature Importance for XGBoost

Key predictors are remain area, year and nepal country .

Attempted feature reduction based on importance scores.

Retained full feature set as removal didn't improve accuracy, suggesting complex feature interactions.

7. Issues Encountered and Proposed Solutions

Outliers in the Dataset

- We identified potential outliers using Mahalanobis Distance.
- However, we did not remove them, as there was no strong evidence to justify their exclusion.

Computational Challenges

- Hyperparameter tuning was time-consuming due to the large dataset (50,000 observations).
- The size of the dataset also highlighted the **need for significant computational power** to handle model training and tuning efficiently.

Low Predictive Performance

- We observed **low accuracy and F1 scores**.
- This may be due to the **weak or no correlation between some predictor variables and the response variable**.
- Additionally, **some variables appeared to be irrelevant** to the outcome, which may have introduced noise into the model.

8. Discussion and Conclusions

Model Performance Comparison

Model	Accuracy		F1 Score	
	Train	Test	Test	
			Low	High
Logistic (FAMD)	0.5052	0.5049	0.4700	0.5400
Random Forest (FAMD)	0.5292	0.5077	0.5100	0.5000
XG Boost	0.5755	0.5107	0.4586	0.5536

Table 5 - Model Performance Comparison

After implementing parameter tuning, factor analysis, dimension reduction and feature selection methods none of the models were able to improve beyond the accuracy level of approximately 50%. Which indicated no model was able to predict the target class significantly better than a random guessing. Alternative targets such as “Yield Grade” were tested and results were merely the same.

We believe that the data does not carry enough predictive information related to the target variables.

However, we were able to identify interacting and latent features through the FAMD.

9. Appendix including R code and technical details.

Codes :
https://drive.google.com/drive/folders/1WjpL2YGYtVX0D579K39wNP1QoO0R3bKv?usp=drive_link

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